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A Kokkos-accelerated Moment Tensor Potential implementation for LAMMPS

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ABSTRACT

We present a Kokkos-accelerated implementation of the Moment Tensor Potential (MTP) for LAMMPS, designed to improve both computational performance and portability across CPUs and GPUs. This package introduces an optimized CPU variant—achieving up to 2× speedups over existing implementations—and two new GPU variants: a thread-parallel version for large-scale simulations and a block-parallel version optimized for smaller systems. It supports three core functionalities: standard inference, configuration-mode active learning, and neighborhood-mode active learning. Benchmarks and case studies demonstrate efficient scaling to million-atom systems, substantially extending accessible length and time scales while preserving the MTP's near-quantum accuracy and native support for uncertainty quantification.

Metadata

Code metadata.

Nr.	Code metadata description	Metadata
C1	Current code version	v1.0.0
C2	Permanent link to code/repository used for this code version	https://github.com/RichardZJM/lammps-mtp-kokkos
C3	Permanent link to Reproducible Capsule	
C4	Legal Code License	MIT License
C5	Code versioning system used	git
C6	Software code languages, tools, and services used	C++, MPI, Kokkos
C7	Compilation requirements, operating environments & dependencies	LAMMPS (24 March 2022 or later)
C8	If available Link to developer documentation/manual	
C9	Support email for questions	contact@richardzjm.com

1. Motivation and significance

Atomistic simulations have become a vital complement to experimental methods in materials discovery and characterization [1–3]. These simulations often rely on interatomic potentials—models of atomic interactions—to compute energies and forces. Traditional potentials are computationally efficient but often lack the flexibility needed for high-fidelity predictions, while quantum mechanical methods such as density functional theory (DFT) offer greater accuracy at substantially higher computational cost. Machine learning interatomic potentials (MLIPs) have emerged as a compelling alternative, providing a systematic framework to improve accuracy by increasing model complexity,

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thereby enabling better control over the cost-accuracy trade-off than conventional approaches [4,5]. However, the computational demands of high-capacity MLIPs remain a challenge, especially for large-scale or long-timescale simulations.

The rise of MLIPs has paralleled growing demand for computational power, making hardware accelerators—particularly GPUs—essential for large-scale simulations. At the same time, the landscape of high-performance computing (HPC) has become increasingly heterogeneous. Leading supercomputers such as LUMI, Frontier, and the newly commissioned El Capitan employ AMD Instinct GPUs (MI250X, MI300A) [6], while Aurora relies on Intel Max GPUs. However, fully leveraging these diverse architectures often requires adopting distinct programming models, vendor-specific frameworks, or low-level languages, posing a significant challenge for portability and maintainability.

LAMMPS—Large-scale Atomic/Molecular Massively Parallel Simulator—a widely used classical atomistic simulation engine [7,8], supports Kokkos-based acceleration to enable performance portability across diverse hardware platforms. Kokkos provides a unified abstraction for parallel execution and data management [9,10], allowing a single implementation to target CPUs, NVIDIA GPUs, AMD GPUs, and other architectures without sacrificing performance. Several interatomic potentials—such as the Tabulated Gaussian Approximation Potential (tabGAP) [11,12], the Spectral Neighbor Analysis Potential (SNAP) [13], and the Atomic Cluster Expansion (ACE/PACE) [14,15]—have been implemented with this framework, and have demonstrated scalability to billion-atom systems and nanosecond timescales [16]. These capabilities bring large-scale, high-fidelity simulations closer to experimentally relevant conditions in both space and time.

We extend the LAMMPS Kokkos package to support the Moment Tensor Potential (MTP)[17], including both inference and uncertainty quantification for active learning. The MTP is one of the most widely used machine learning interatomic potentials, with demonstrated success across metals, semiconductors, and multicomponent systems [18–31]. It offers a strong balance of computational efficiency, systematic improvability, and built-in support for active learning with extrapolation grades based on D-optimality [32] and the MaxVol algorithm [33].

A recent study has shown that ACE outperforms the MTP in two test cases on Pareto fronts of accuracy versus cost [34], although another study has shown the MTP to be more performant in their use case [35]. However, since one can be cast in the form of the other, the accuracy can be similar [36,37]. The MTP is still undergoing development, which has shown that careful adjustments to the basis construction can further improve its performance and expressiveness [25], a direction we are also exploring. Integrating the MTP into the Kokkos framework within LAMMPS enables scalable, portable deployment across GPU architectures, significantly broadening the range of accessible simulation sizes and timescales.

The software is available on GitHub[1] [38] and supports LAMMPS[2] versions newer than the 24 March 2022 version and will be maintained to work with future LAMMPS stable releases.

2. Software description

2.1. Software architecture

We implement nine new variations of the Moment Tensor Potential, each optimized for different simulation conditions to provide flexibility across a wide range of system sizes and hardware configurations. These cover three core use cases: inference, active learning in configuration mode, and active learning in neighborhood mode. For each use case, we provide three implementations: (1) a further optimized CPU version (non-Kokkos) that improves upon the previous MLIP-3 package [39]; (2) a thread-parallel GPU variant designed for large-scale simulations (typically ≳50,000 atoms per GPU); and (3) a block-parallel GPU variant optimized for smaller simulations (typically ≳2000 atoms per GPU). The block-parallel version, denoted with `small` in the LAMMPS `pair_style`, exposes additional fine-grained parallelism, which can improve performance at small-to-intermediate system sizes but may reduce peak throughput. Each variant follows LAMMPS `pair_style` naming conventions, as summarized in Table 1.

Table 1
Each MTP variation and its LAMMPS `pair_style` identifier.

	Platform	Use Cases	
		Inference	Active Learning, Both Modes
CPU		`mtp`	`mtp/extrapolation`
GPU	Thread-Parallel	`mtp/kk`	`mtp/extrapolation/kk`
	Block-Parallel	`mtp/small/kk`	`mtp/extrapolation/small/kk`

Users should first select the desired use case. Inference computes energies, forces, and stresses during standard molecular dynamics simulations. Active learning performs the same calculations while also evaluating the extrapolation grade at user-defined intervals, enabling on-the-fly model improvement. Configuration mode and neighborhood mode refer to two distinct strategies for computing this grade. As in MLIP-3, both modes support a selection threshold, above which configurations are written to a file, and a break threshold, above which the simulation is halted. Once the use case is chosen, the appropriate implementation (CPU, thread-parallel GPU, or block-parallel GPU) can be selected. For GPU usage, we recommend short single-GPU trial runs of both variants, as relative performance depends strongly on the MTP hyperparameters and the hardware, especially if consumer GPU hardware is used. These performance variations are part of the reason we offer two different styles of GPU parallelism.

2.2. Software functionalities

The main contributions of the software are improved CPU performance and new GPU capabilities, while fully preserving the core functionality of the MTP as described in previous works [17,39]. To evaluate these improvements, we benchmarked both weak and strong scaling on the Digital Research Alliance of Canada's Narval HPC cluster. Narval's CPU nodes are equipped with 2× AMD EPYC™ 7532 processors (32 cores each), and its GPU nodes, with 4× NVIDIA A100 SXM4 (40 GB). MTP models are characterized by their "level", which exponentially scales the number of basis functions, and thus the computational cost. Fig. 1 presents weak and strong scaling results across a range of MTP levels, using a quarter, half, and full Narval node (CPU and GPU). All benchmarks use default hyperparameters (e.g., cutoff radius) for a bulk simulation of unstrained solid potassium in NVE. The complete LAMMPS input script and full benchmarking data are provided in a separate repository [40]. Importantly, the performance of the algorithm is heavily dependent on the hardware, the material, and the simulation conditions. The same throughput should not be expected in all simulations.

To mitigate GPU memory constraints, the GPU implementations employ a `chunksize` parameter. This parameter breaks atom processing into chunks for each timestep, allowing the reuse of working memory and enabling larger systems without significantly affecting throughput. For simplicity, Fig. 1 presents measurements exclusively from simulation rates attainable with a single chunk.

For each MTP level, we evaluated the maximum throughput across all the tested atom counts and trials, and reported the relative speedups. Inference speedups compared to the original MLIP-3 implementation are shown in Fig. 2, using a single CPU core and a single NVIDIA A100 GPU.

[1] https://github.com/RichardZJM/lammps-mtp-kokkos.
[2] LAMMPS' default CPU architecture is 64-bit x86 and 64-bit ARM is regularly tested.

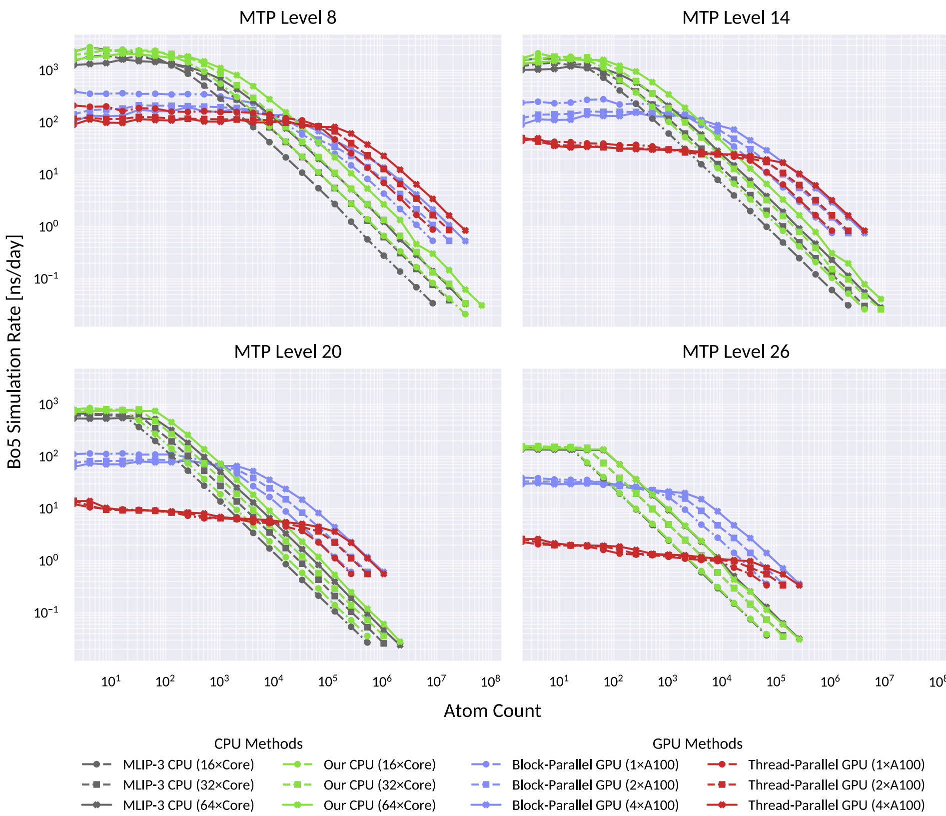

Fig. 1. A log-log plot of the inference simulation rate versus the atom count of several MTP implementations on various hardware for selected MTP levels. Separate simulations are performed for 100 timesteps for each atom count and method, and the best of five (Bo5) simulation rate is reported. A 1 fs timestep is used. CPU measurements were taken until a rate of less than 0.02 ns/day was observed. GPU measurements were taken until there was insufficient memory to compute all atoms in a single chunk. Multiple chunks can be used to handle larger problems at a similar throughput.

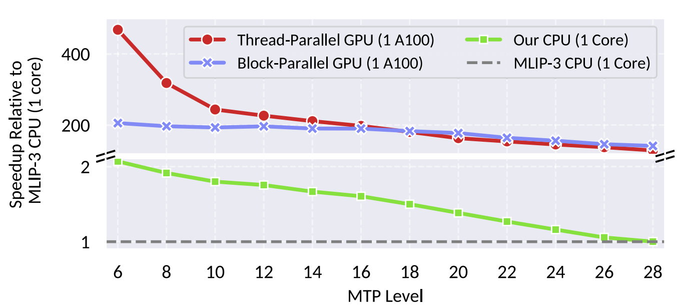

Fig. 2. The inference speedups (relative maximum throughput) over the previous MLIP-3 implementation.

Fig. 3 presents the corresponding speedups for active learning in both configuration and neighborhood modes, evaluated across simulations with one, two, or three atomic species.[3]

Notably, the crossover point at which the thread-parallel variant outperforms the block-parallel variant depends strongly on both the atom count and MTP level, as well as the underlying hardware. Overall, the observed GPU speedups are comparable to the acceleration achieved by the existing Kokkos implementation of the ACE potential when comparing a single A100 GPU to a single CPU core [34].

[3] The simulation still uses only potassium, although the potentials make extrapolation grade calculations as if there were more species.

Details about installing the implementations and invoking them in a LAMMPS script are available in the Supplementary Material and are documented in the code repository [38].

3. Illustrative examples

We demonstrate the GPU variants in three illustrative examples where GPUs are helpful: (1) a large simulation with a high MTP level, (2) a very large simulation with a medium MTP level, and (3) a medium-scale simulation with active learning enabled. These examples also target important phenomena, specifically capturing dislocation glide in silicon, grain boundary mechanics in nanocrystalline aluminum, and the melting point in eutectic sodium-potassium. Table 2 summarizes the simulation characteristics and rates achieved by MLIP-3 (on 64 cores) and our GPU methods. Detailed descriptions and visualizations are available in the Supplementary Material. The corresponding LAMMPS input scripts for these case studies are available in the external data repository [40].

4. Impact

When exploring, discovering, and characterizing materials through atomistic simulations, many phenomena of interest require simulations involving millions of atoms. Examples include amorphous materials [41], crack propagation [42], nanocrystalline systems [43], irradiation damage [44], and dislocation dynamics and plasticity [45]. Certain materials, such as multi-principal element alloys [35,46], oxide dispersion-strengthened alloys [47], nanocomposites [48], and metallic glasses [49] also require very large simulations to study.

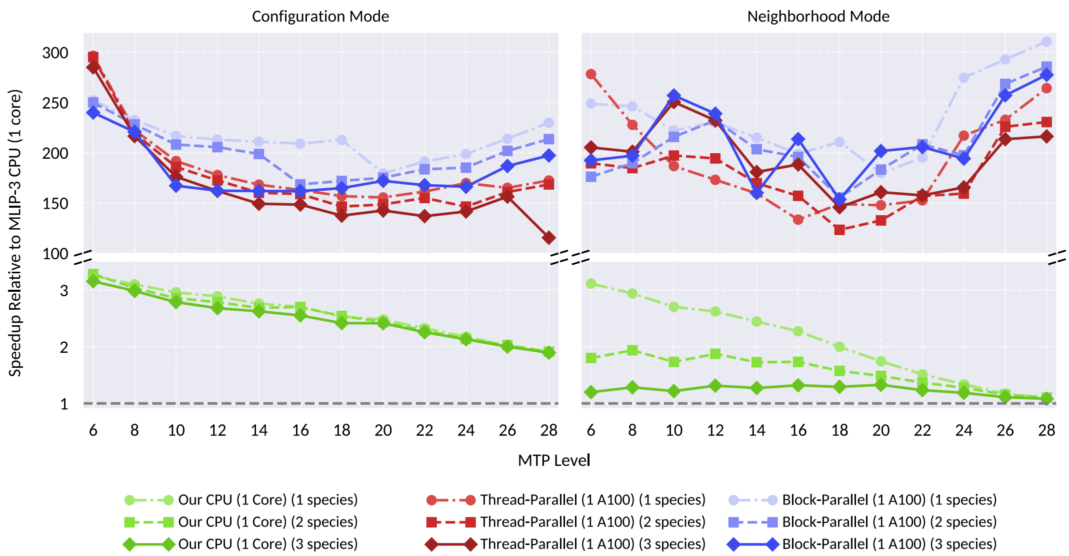

Fig. 3. The active learning speedups (relative maximum throughput) for both configuration and neighborhood mode over the previous MLIP-3 implementation.

Table 2
A table summarizing the three illustrative examples and the rates achieved by MLIP-3 and the GPU implementations. In example 3, MLIP-3 exhibited worse strong scaling, leading to the larger performance difference.

	1. Dislocation Glide	2. Nanocrystalline Tension	3. Active Learning & Melting
Simulation Type	Large scale, high MTP level.	Very large scale, medium level.	Medium scale with active learning.
System	Silicon Screw Dislocation	Nanocrystalline Aluminum	Na–K Solid–Liquid Interface
Atom Count	115 thousand	1.00 million	3600
MTP Level	26	16	18
MLIP-3 (64 cores)	**0.032 ns/day**	**0.026 ns/day**	**0.535 ns/day**
GPU Rate	**0.515 ns/day**	**0.453 ns/day**	**14.925 ns/day**
Variant	Block-Parallel	Thread-Parallel	Block-Parallel
GPUs	4×A100	4×A100	1×A100

By introducing a family of GPU-accelerated variants, we significantly expand the MTP's applicability to larger, more complex simulations. These implementations enable faster time-to-solution for many existing problems and make it feasible to deploy higher-level, more accurate MTPs that would otherwise be computationally prohibitive. The GPU implementations also enable medium-scale simulations on consumer-grade hardware or with Multi-Instance GPU (MIG), improving accessibility across a broader range of research environments.

This software contribution is part of a collective effort to improve upon the MTP and other similar potentials, such as the Equivariant Tensor Network Potential, the latter of which could be enhanced to support Kokkos.

5. Conclusions

Despite its popularity, demonstrated successes, native support for active learning, and strong computational efficiency, the MTP has previously lacked GPU support—limiting its scalability on modern HPC systems. We have addressed this limitation by introducing a family of GPU-accelerated variants alongside an optimized CPU implementation that outperforms the original implementation, with speedups of up to 2×.

CRediT authorship contribution statement

Zijian Meng: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Karim Zongo:** Writing – review & editing, Validation. **Edmanuel Torres:** Writing – review & editing. **Christopher Maxwell:** Writing – review & editing. **Ryan Grant:** Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition. **Laurent Karim Béland:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Gemini to assist in writing plotting scripts in Python for benchmark data. After using this tool, the authors reviewed and edited the plots as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data for this article can be found online at doi:10.1016/j.softx.2026.102524.

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